

Digital Logic Design

Lecture 1

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Digital Logic and AI

- **Digital logic** and **AI** are interconnected fields where digital logic forms the foundational hardware infrastructure that enables **AI** systems to function effectively.
- **Digital logic** involves designing the electronic circuits and components that process binary signals, which are essential for building the processors, memory, and specialized hardware used in **AI** applications.
- **AI** relies on these digital systems to perform complex computations, run algorithms, and handle large data sets efficiently.
- In AI Engineering, digital logic circuits are used to develop the processors, memory units, and specialized hardware such as **neural network accelerators** and **AI chips** that enable high-speed data processing and complex computations.
- Overall, digital logic provides the hardware backbone that supports the development, deployment, and scalability of **AI** technologies.

Digital Logic Design Relationship to AI Engineering

- Digital Logic Design is fundamentally related to AI Engineering as it provides the essential foundation for building the hardware that supports artificial intelligence systems.
- Understanding digital logic design allows AI engineers to optimize hardware performance, improve energy efficiency, and develop custom digital components tailored for AI algorithms.
- Additionally, the principles of digital logic are crucial for designing embedded systems and hardware accelerators that execute AI models in real-time, making digital logic design a critical underlying technology for advancing AI hardware infrastructure.

Digital Logic Design Definition

- Digital Logic Design is the process of creating and analyzing digital circuits that perform logical operations using discrete signals, primarily binary signals (0s and 1s).
- It involves designing systems that manipulate these signals to implement functions such as **addition, subtraction, data storage, and decision-making**.
- The field combines principles of **electronics** and **Boolean algebra** to develop reliable, efficient digital components and systems, which form the foundation of all digital devices like computers, microprocessors, and digital systems.

Digital Computers and Digital Systems

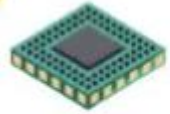
- **Computer:** The digital computer consists of three main units:
 - **Memory Unit:** Stores programs, input, output, and intermediate data.
 - **Processor Unit:** Performs arithmetic and data-processing tasks as specified by the program.
 - **Control Unit:** Supervises the flow of information, retrieves instructions from memory, and directs the processor to execute them.
- **Program and Data Storage:** Both program instructions and data are stored in the memory unit. The control unit manages the execution of instructions, while the processor manipulates the data.
- **Input and Output Devices:** Input devices (e.g., mouse, keyboard, etc) transfer user-prepared programs and data into memory. Output devices (e.g., printer) present computation results to the user.
- **These devices are special digital systems controlled by electronic digital circuits.**

Parts of Computer



Q mrmrsEnglish

Search



CPU



Webcam



HDD



SSD



GPU



Microphone



Speakers



Motherboard



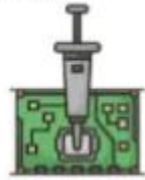
Monitor



External Drive Bays



SATA Connectors



Thermal Paste



VRM



Internal Drive Bays



RAID Controller



Modem



M.2 Slot



RAM



PSU



Optical Drive



Sound Card



NIC



Wi-Fi Card



Cooling Fans



Heatsin



CMOS Battery



Expansion Slots



BIOS



Ports (USB)



I/O Shield



Case or Tower



Keyboard



Mouse



Chipset



USB Hub



Bluetooth Adapter

Digital Computers and Digital Systems

Comparison with Electronic Calculators:

Electronic calculators are similar to digital computers but are less powerful. Calculators use a keyboard for input and a numeric display for output, and some have limited programmability. Digital computers, however, can handle a wider range of input/output devices, perform both arithmetic and logical operations, and make decisions based on conditions.



- **CPU and Microprocessor:**

- **CPU (Central Processor Unit):** Combination of processor and control unit.
- **Microprocessor:** A CPU in a small integrated-circuit package. May also include memory and interface control.
- **Microcomputer:** A small computer system combining CPU, memory, and interface control, often in a single package. **The rise of microcomputers has revolutionized digital system design.**
- **Binary Representation:** Digital computers manipulate discrete information represented in binary form. Calculations use binary numbers, and other data (like decimal digits) are encoded in binary codes. **Data processing uses binary logic elements and signals, and storage is in binary storage elements.**

To Begin with Digital Logic Understanding, We need to Know:

- Number Systems
- Types of Numbers Systems
- Logic gates
- Boolean algebra

Number Systems in Digital Logic

- A decimal number such as **7392** represents a quantity equal to 7 thousands plus 3 hundreds, plus 9 tens, plus 2 units. The thousands, hundreds, etc., are powers of 10 implied by the position of the coefficients. To be more exact, 7392 should be written as:

$$7 \times 10^3 + 3 \times 10^2 + 9 \times 10^1 + 2 \times 10^0$$

- In general, a number with a decimal point is represented by a series of coefficients as follows:

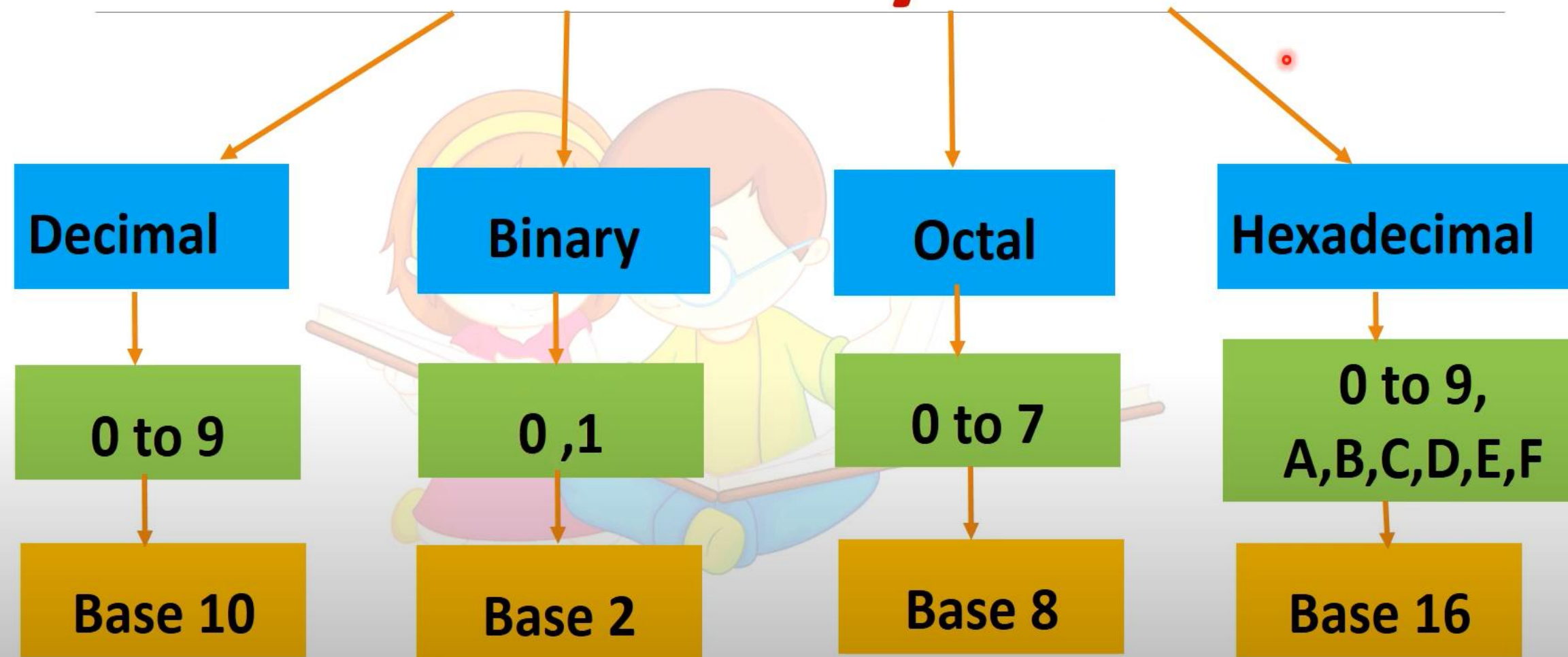
$$10^5 a_5 + 10^4 a_4 + 10^3 a_3 + 10^2 a_2 + 10^1 a_1 + 10^0 a_0 + 10^{-1} a_{-1} + 10^{-2} a_{-2} + 10^{-3} a_{-3}$$

- The decimal number system is said to be of *base*, or *radix* 10 because it uses ten digits and the coefficients are multiplied by powers of 10

Number Systems in Electronics

- The technique to represent and work with numbers is called **number system**
- **Two types of number systems we have:**
 1. Non-computer number system e.g. decimal number system. Decimal number system has only ten (10) digits from **0 to 9**. Every number (value) represents with 0,1,2,3,4,5,6,7,8 and 9 in this number system. The base of decimal number system is 10, because it has only 10 digits.
 2. Computer number system, also called **digital number system**. For example, binary number system, octal number system and hexadecimal number system.

Number system



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graph TD; A[Number system] --> B[Decimal]; A --> C[Binary]; A --> D[Octal]; A --> E[Hexadecimal]; B --> F[0 to 9]; F --> G[Base 10]; C --> H[0, 1]; H --> I[Base 2]; D --> J[0 to 7]; J --> K[Base 8]; E --> L["0 to 9, A, B, C, D, E, F"]; L --> M[Base 16];
```

Decimal

0 to 9

Base 10

Binary

0, 1

Base 2

Octal

0 to 7

Base 8

Hexadecimal

**0 to 9,
A, B, C, D, E, F**

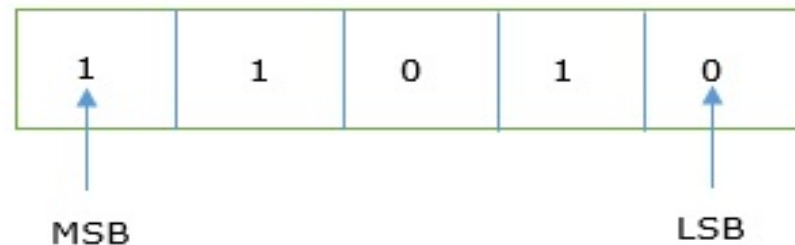
Base 16

Digital (Computer) Number System

Binary Number System.

- A Binary number system has only two digits that are **0 and 1**.
- Each binary digit is also called a **bit**.
- The base of binary number system is 2, because it has only two digits.
- Binary number system is a positional value system, where each digit has a value expressed in powers of 2.

In any binary number, the rightmost digit is called **least significant bit (LSB)** and leftmost digit is called **most significant bit (MSB)**.



Binary Number Representation

Binary Value	Decimal Representation				Decimal Value			
	8	4	2	1				
0 0 0 0	0	+	0	+	0	+	0	0
0 0 0 1	0	+	0	+	0	+	1	1
0 0 1 0	0	+	0	+	2	+	0	2
0 0 1 1	0	+	0	+	2	+	1	3
0 1 0 0	0	+	4	+	0	+	0	4
0 1 0 1	0	+	4	+	0	+	1	5
0 1 1 0	0	+	4	+	2	+	0	6
0 1 1 1	0	+	4	+	2	+	1	7
1 0 0 0	8	+	0	+	0	+	0	8
1 0 0 1	8	+	0	+	0	+	1	9
1 0 1 0	8	+	0	+	2	+	0	10

Example

To find the decimal number equivalent to the binary number $(11010)_2$

$$\begin{aligned} 11010_2 &= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \\ &= 16 + 8 + 0 + 2 + 0 \\ &= 26_{10} \end{aligned}$$

Digital (Computer) Number System

- Octal number system

Octal number system has only eight (8) digits from **0 to 7**. Every number (value) represents with 0,1,2,3,4,5,6 and 7 in this number system. The base of octal number system is 8, because it has only 8 digits.

8^5	8^4	8^3	8^2	8^1	8^0
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Example

- Decimal equivalent of any octal number is sum of product of each digit with its positional value.

$$726_8 = 7 \times 8^2 + 2 \times 8^1 + 6 \times 8^0$$

$$= 448 + 16 + 6$$

$$= 470_{10}$$

Counting in Octal Number System

Decimal	Binary	Octal
0	000	0
1	001	1
2	010	2
3	011	3
4	100	4
5	101	5
6	110	6
7	111	7
8	001 000	10
9	001 001	11
10	001 010	12
11	001 011	13
12	001 100	14

Digital (Computer) Number Systems

- Hexadecimal number system

A Hexadecimal number system has sixteen (16) alphanumeric values from **0 to 9** and **A to F**. Every number (value) represents with 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E and F in this number system. The base of hexadecimal number system is 16, because it has 16 alphanumeric values. Here **A is 10**, **B is 11**, **C is 12**, **D is 13**, **E is 14** and **F is 15**.

16^5	16^4	16^3	16^2	16^1	16^0
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Example

- Decimal equivalent of any hexadecimal number is sum of product of each digit with its positional value.

$$\begin{aligned}(27FA)_{16} &= 2 \times 16^3 + 7 \times 16^2 + 15 \times 16^1 + 10 \times 16^0 \\ &= 8192 + 1792 + 240 + 10 \\ &= (10234)_{10}\end{aligned}$$

HEXADECIMAL	DECIMAL	OCTAL	BINARY
0	0	0	0000
1	1	1	0001
2	2	2	0010
3	3	3	0011
4	4	4	0100
5	5	5	0101
6	6	6	0110
7	7	7	0111
8	8	10	1000
9	9	11	1001
A	10	12	1010
B	11	13	1011
C	12	14	1100
D	13	15	1101
E	14	16	1110
F	15	17	1111